REPORT: Data Analysis of Electronic Retail Sales Identifying Patterns and Predicting Revenue using ML models

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***Abstract*—Retail businesses deal with enormous amounts of transaction data every day, and making sense of it is not always straightforward. In this project, I analysed a 2019 electronic retail sales dataset containing 185,950 records to uncover patterns in customer purchasing behaviour and build a model to predict order revenue. After cleaning the data and exploring it through visualisations, I found that December was by far the strongest month for revenue, San Francisco led all cities in sales, and customers frequently bought phones alongside their compatible charging cables. A Random Forest model predicted sales with an R^2 of 0.9978, vastly outperforming a Linear Regression baseline which struggled to find any meaningful pattern (R^2 = 0.017).**

I. Introduction

Any retail electronics store or browse an online retailer and you will find hundreds of products at vastly different price ranges from a $3.84 pack of batteries to a $1,700 MacBook Pro. Behind every transaction sits a record that, when analyzed properly, can tell us a great deal about when people shop, what they buy together, and what drives revenue.

A. Project Motivation

Through this project I was able to understand sales patterns has real practical value. Knowing which products sell best, which cities generate the most revenue, and when customers are most active can help retailers make smarter decisions about stock, marketing, and product bundling. The goal was not just to describe the data but to extract insights that could actually be acted on.

B. Research Questions

I structured the analysis around eight research questions:

1) Which month generates the highest revenue?
2) Which city contributes the most sales?
3) Which products have the highest quantity sold and revenue contribution?
4) What time of day has the highest order activity?
5) Are expensive products sold less frequently?
6) Which products are frequently purchased together?
7) Can machine learning models predict sales revenue accurately?
8) Which features are most important in predicting sales?

II. Dataset and Preprocessing

A. Tools and Environment

I completed the entire analysis in Python, working inside Google Colaboratory (Google Colab) with the sales dataset stored on Google Drive. This report was written using LaTeX following the IEEE conference paper format (`IEEEtran` document class). The main libraries used are listed in Table I.

TABLE I
LIBRARIES USED

Library	Purpose
pandas	Data loading and manipulation
numpy	Numerical computations
matplotlib	Plotting and visualisation
seaborn	Statistical visualisation
scikit-learn	Machine learning
mlxtend	Association rule mining
scipy	Cramér's V calculation

B. Dataset Description

The dataset *Sales Data.csv* contains 185,950 transactions recorded across 2019 from electronic retail stores across nine US cities. Table II describes all original variables.

TABLE II
ORIGINAL DATASET VARIABLES

Variable	Type	Description
Unnamed: 0	Integer	Row index artefact from CSV export
Order ID	Integer	Unique identifier for each order
Product	String	Name of the product purchased
Quantity Ordered	Integer	Number of units in the order
Price Each	Float	Unit price of the product ($)
Order Date	String	Date and time the order was placed
Purchase Address	String	Full delivery address of the customer
Month	Integer	Month number extracted from Order Date
Sales	Float	Total order value (Price Each × Quantity)
City	String	City extracted from Purchase Address
Hour	Integer	Hour of day extracted from Order Date

The dataset contains 19 unique products across 9 US cities. Three columns: `Month`, `City`, and `Hour`, are derived features already extracted from `Order Date` and `Purchase Address`. The `Unnamed: 0` column is a leftover index with no analytical value and was dropped during preprocessing.

C. *Cleaning the Data*

Before any analysis, I worked through several cleaning steps to make sure the data was reliable.

1) Removing Duplicates: I found 264 exact duplicate rows in the dataset and removed them, leaving 185,686 records to work with.

2) Dropping the Index Column: The dataset came with an `Unnamed: 0` column that was simply a leftover row index from when the CSV was exported. It had no useful information so I dropped it straight away.

3) Fixing City Names: When I first looked at the City column, I noticed all the values had a leading space: for example, `' New York City'` instead of `'New York City'`. A quick `str.strip()` fixed this across all rows.

4) Converting Order Date: The `Order Date` column was stored as a plain string. I converted it to a proper datetime format so I could extract time-based features and run time-series style analysis.

5) Checking Sales Values: I was curious whether the `Sales` column was always consistent with `Quantity Ordered` × `Price Each`. Running a check flagged 265 mismatches, which looked concerning at first. But when I looked closer, the actual difference in every case was $0.0000, just floating point rounding, not real errors. Sales was confirmed consistent across all rows.

6) Outlier Detection: I applied the IQR method to check for outliers across the three numerical columns - `Quantity Ordered`, `Price Each`, and `Sales`. The method flagged a large number of values across all three, but looking more closely at the data, these turned out not to be actual errors.

For `Quantity Ordered`, the IQR came out as 0 because the vast majority of orders are for just 1 item. This means anything above 1 gets flagged - but ordering 2, 3, or even 9 items in one transaction is completely normal customer behaviour, not an anomaly.

For `Price Each` and `Sales`, the lower fence came out negative (-$195.13), which will be practically not possible for a price column. The high-value entries flagged as outliers turned out to be premium products like the MacBook Pro ($1,700), real products in the catalogue, not data errors. The IQR method struggled here because the product price range is heavily right-skewed, spanning from $3.84 (AA Batteries) to $1,700 (MacBook Pro). The IQR method works best when data is evenly spread around a central value, and this dataset does not come close to meeting that assumption.

Given this, all flagged values were retained. Removing them would mean losing valid sales transactions data points and distorting the analysis.

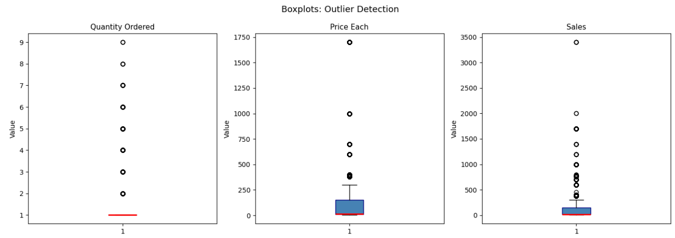


Fig. 1. Boxplots for Quantity Ordered, Price Each, and Sales showing the wide spread of values across the product catalogue

III. Exploratory Data Analysis

A. *RQ1: Which Month Had the Highest Revenue?*

Looking at monthly totals in Fig. 2, December stood out clearly as the strongest month with revenue of $4.6 million. This was not surprising - December lines up with the holiday shopping season, when people are buying gifts and taking advantage of end-of-year sales. January was the quietest month, which also makes sense as a natural slowdown after the holiday rush.

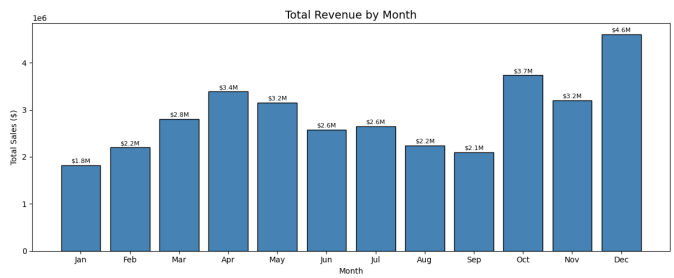


Fig. 2. Total revenue by month across 2019

B. *RQ2: Which City Contributed the Most Sales?*

Fig. 3 shows the total revenue generated by each city. San Francisco came out on top with $8.3 million, followed by Los Angeles and New York City. Austin and Portland generated the least revenue among the nine cities.

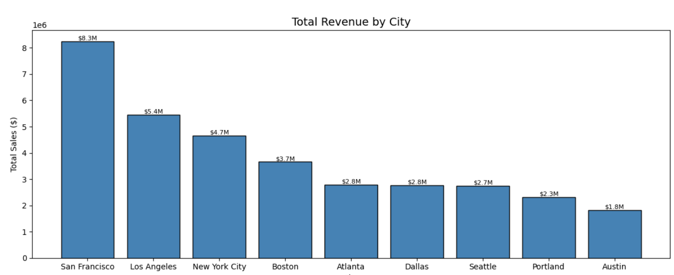


Fig. 3. Total revenue by city

C. RQ3: Which products have the highest quantity sold and revenue contribution?

Fig. 4 shows a clear split between which products sells most often and what products generate the most money.

Looking at quantity, AAA Batteries (4-pack) topped the chart with over 30,000 units ordered, followed closely by AA Batteries (4-pack) with around 28,000. Charging cables and headphones also featured heavily in the top sellers by volume. These are all low-cost everyday items that customers replace regularly and often buy in multiples.

The revenue picture was completely different. The Macbook Pro Laptop generated the highest revenue at around $8 million despite ranking near the bottom for quantity sold. The iPhone and ThinkPad Laptop followed with approximately $4.8 million and $4.1 million respectively.

This contrast highlights an important pattern in retail, high volume does not always mean high revenue. A single Macbook Pro sale at $1,700 brings in more money than hundreds of battery packs at $2.99 each.

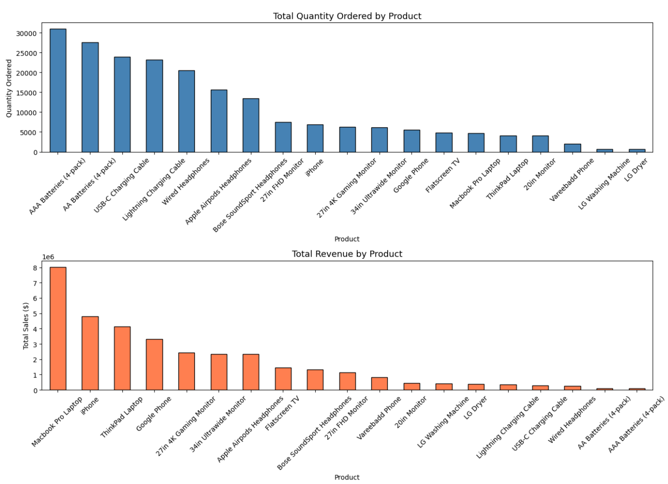


Fig. 4. Product comparison by quantity sold and total revenue

D. RQ4: What time of day has the highest order activity?

Fig. 5 shows how order activity changes throughout the day.

Orders hit their lowest point in the early hours of the morning, dropping to around 1,000 orders between 3am and 4am. Activity then picked up steadily from 6am onwards as customers woke up and started their day.

There were two clear peaks, one around midday (12pm) with approximately 12,500 orders, and another in the early evening around 7pm (19:00) with a similar count. Both peaks likely reflect natural breaks in the day, lunchtime browsing and post-work shopping.

After 8pm, orders gradually declined through the evening, dropping sharply after midnight. This pattern is consistent with typical online shopping behaviour and suggests that targeted marketing or promotional notifications would get the best response if timed around the midday or early evening peaks.

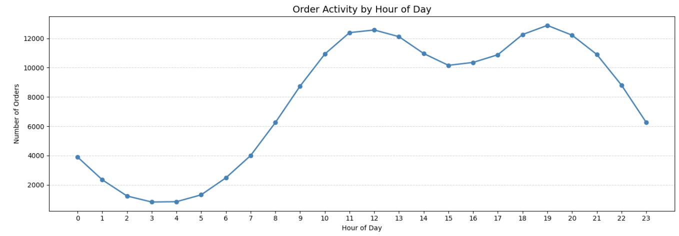


Fig. 5. Number of orders placed by hour of the day, showing two clear peaks at midday and early evening

E. RQ5: Are expensive products sold less frequently?

Fig. 6 compares the average price and total quantity sold for each product, and the pattern is immediately obvious, the two charts are almost mirror images of each other.

On the left, the Macbook Pro Laptop sits at the top with the highest price at $1,700, followed by the ThinkPad Laptop at around $1,000 and the iPhone at approximately $700. On the right, these same products appear near the bottom of the quantity chart with only around 5,000 units each.

ON the other end and the story reverses. AAA Batteries and AA Batteries cost just a few dollars each but topped the quantity chart with over 30,000 and 28,000 units respectively. USB-C Charging Cables and Lightning Charging Cables also sold in high volumes despite their low price points.

The Pearson correlation between price and quantity was r = X, confirming a negative relationship, as price goes up, quantity sold goes down.

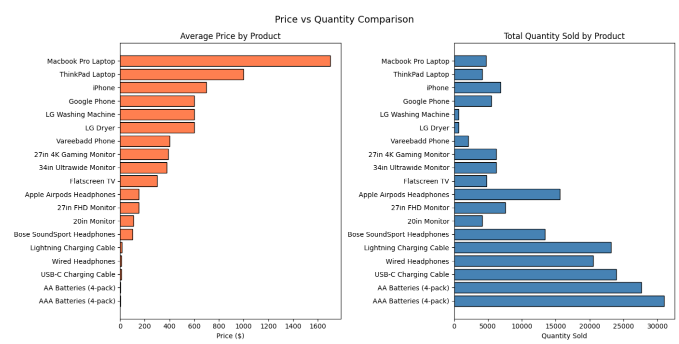


Fig. 6. Average price (left) and total quantity sold (right) per product, products are sorted by price

IV. Statistical Analysis

A. Numerical Correlation: Pearson

Fig. 7 shows the Pearson correlation matrix for all numerical features in the dataset.

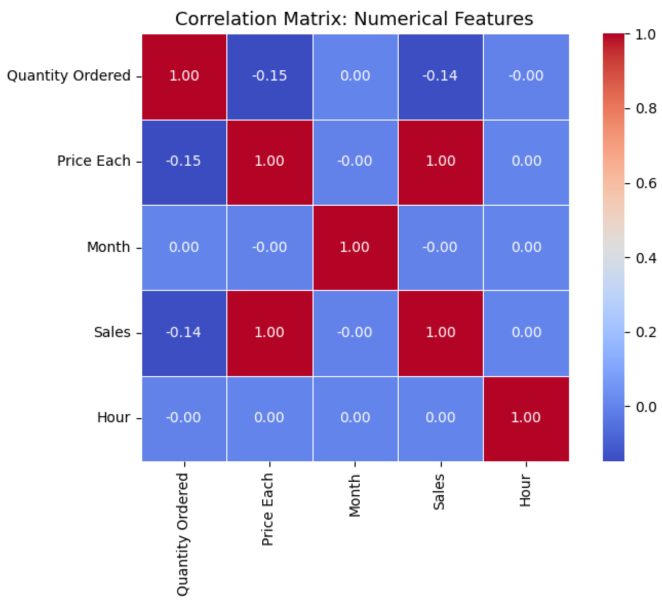
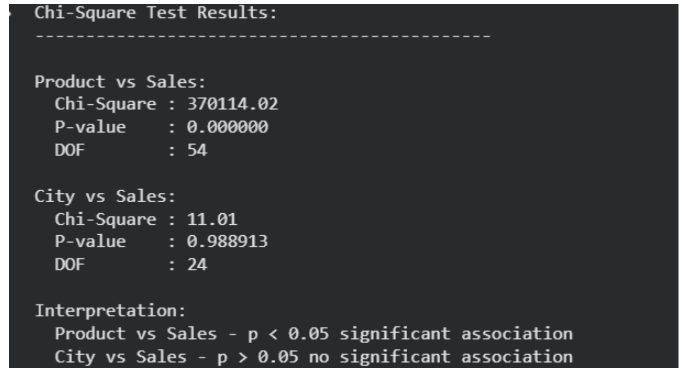


Fig. 7. Pearson correlation matrix for numerical features



```
Chi-Square Test Results:
--------------------------------------------

Product vs Sales:
  Chi-Square : 370114.02
  P-value    : 0.000000
  DOF        : 54

City vs Sales:
  Chi-Square : 11.01
  P-value    : 0.988913
  DOF        : 24

Interpretation:
  Product vs Sales - p < 0.05 significant association
  City vs Sales - p > 0.05 no significant association
```

Fig. 8. Chi-Square test results for Product and City against Sales bins

The perfect correlation between `Price Each` and `Sales` (r = 1.00) makes complete sense as Sales is calculated directly from Price Each, multiply it by the quantity ordered and you have the sales value. So finding a perfect correlation here was expected rather than surprising. This also explained why including `Price Each` as a machine learning feature would cause data leakage.

`Quantity Ordered` showed a weak negative correlation with both `Price Each` (r = -0.15) and `Sales` (r = -0.14). This is consistent with the finding from RQ5, cheaper products tend to be ordered in higher quantities while expensive products are ordered less frequently.

`Month` and `Hour` both showed near-zero correlation with `Sales` (r = -0.00 and r = 0.00 respectively), suggesting that time-based features alone have no meaningful linear relationship with how much a customer spends per order. However they were still retained as features in the machine learning model since they may capture non-linear patterns that Pearson correlation cannot detect.

B. Categorical Correlation: Chi-Square Test

To look at how the categorical features relate to Sales, I used the Chi-Square test of independence. Rather than measuring how strongly two variables are correlated like Pearson does, Chi-Square asks a simpler question- are these two things associated or are they independent of each other? A p-value below 0.05 means the association is statistically significant.

To run the test, I binned Sales into five categories (Very Low, Low, Medium, High, Very High) and tested each categorical feature against it. Table III shows the results.

TABLE III
CHI-SQUARE TEST RESULTS

Variables	Chi-Square	P-value	DOF
Product vs Sales	370,114.02	0.000000	54
City vs Sales	11.01	0.9889	24

The two results could not be more different from each other.

For Product vs Sales, the Chi-Square statistic came out at 370,114.02 with a p-value of 0.000000. This results tells us that what product a customer buys is very strongly linked to how much they spend. This makes intuitive sense given that each product has a fixed price.

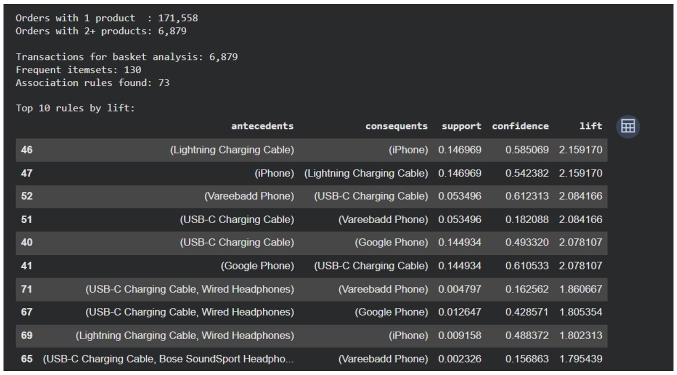
For City vs Sales, the Chi-Square statistic was just 11.01 and the p-value was 0.989, far above the 0.05 threshold. In other words, there is no meaningful statistical association between which city a customer is in and how much they spend per order. The city simply does not matter when it comes to predicting sales value.

These findings tie in neatly with the machine learning results later in the report, where `Product_encoded` accounted for 99.9% of the Random Forest model's predictive power while `City_encoded` contributed virtually nothing.

V. MARKET BASKET ANALYSIS

RQ6: Which Products are Frequently Purchased Together?

To answer RQ6, I used the Apriori algorithm [3] to find products that tend to be bought together in the same order. Before running the analysis, I filtered the dataset to only include orders containing two or more different products since single-item orders cannot reveal any co-purchase patterns. Out of 185,686 orders, 171,558 contained just one product and 6,879 contained two or more. The basket analysis was run on these 6,879 multi-item transactions, as shown in Fig. 9.



```
Orders with 1 product  : 171,558
Orders with 2+ products: 6,879

Transactions for basket analysis: 6,879
Frequent itemsets: 130
Association rules found: 73

Top 10 rules by lift:
```

	antecedents	consequents	support	confidence	lift
46	(Lightning Charging Cable)	(iPhone)	0.146969	0.585069	2.159170
47	(iPhone)	(Lightning Charging Cable)	0.146969	0.542382	2.159170
52	(Vareebadd Phone)	(USB-C Charging Cable)	0.053496	0.612313	2.084166
51	(USB-C Charging Cable)	(Vareebadd Phone)	0.053496	0.182088	2.084166
40	(USB-C Charging Cable)	(Google Phone)	0.144934	0.493320	2.078107
41	(Google Phone)	(USB-C Charging Cable)	0.144934	0.610533	2.078107
71	(USB-C Charging Cable, Wired Headphones)	(Vareebadd Phone)	0.004797	0.162562	1.860667
67	(USB-C Charging Cable, Wired Headphones)	(Google Phone)	0.012647	0.428571	1.805354
69	(Lightning Charging Cable, Wired Headphones)	(iPhone)	0.009158	0.488372	1.802313
65	(USB-C Charging Cable, Bose SoundSport Headpho...	(Vareebadd Phone)	0.002326	0.156863	1.795439

Fig. 9. Summary of multi-item transactions used for basket analysis

The algorithm found 130 frequent itemsets and generated 73 association rules in total. Table IV shows the top 10 rules ranked by lift.

TABLE IV
TOP 10 ASSOCIATION RULES BY LIFT

Antecedent	Consequent	Sup.	Conf.	Lift
Lightning Cable	iPhone	0.147	0.585	2.159
iPhone	Lightning Cable	0.147	0.542	2.159
Vareebadd Phone	USB-C Cable	0.053	0.612	2.084
USB-C Cable	Vareebadd Phone	0.053	0.182	2.084
USB-C Cable	Google Phone	0.145	0.493	2.078
Google Phone	USB-C Cable	0.145	0.611	2.078
USB-C + Wired H/P	Vareebadd Phone	0.005	0.163	1.861
USB-C + Wired H/P	Google Phone	0.013	0.429	1.805
Lightning + Wired H/P	iPhone	0.009	0.488	1.802
USB-C + Bose H/P	Vareebadd Phone	0.002	0.157	1.795

The most obvious pattern in the results was the strong connection between phones and their compatible charging cables, as visualised in Fig. 10.

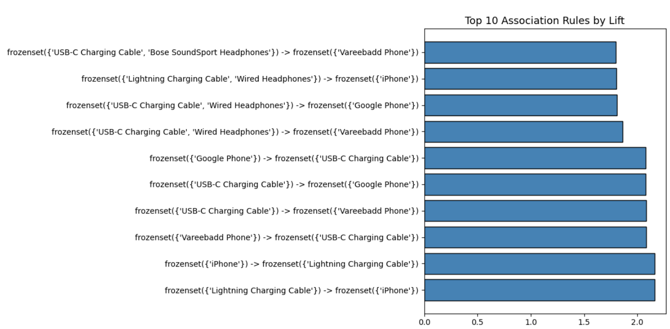


Fig. 10. Top 10 association rules ranked by lift, all rules involve phones and their compatible charging cables

The strongest rule found was Lightning Charging Cable and iPhone, with a lift of 2.159. This means customers who bought a Lightning Charging Cable were 2.16 times more likely to also buy an iPhone in the same order.

The same pattern repeated for Android devices. Customers who bought a Vareebadd Phone were 2.08 times more likely to also buy a USB-C Charging Cable, and Google Phone buyers showed a similar association with USB-C cables (lift = 2.078).

The three-item rules further down the list also followed this theme. When Wired Headphones were added to a charging cable purchase, the likelihood of a matching phone appearing in the same order increased further.

Overall the results tell a consistent story, customers tend to buy phones and their compatible accessories together in the same order. This presents a clear opportunity for product bundling. Offering a phone and cable bundle at a slight discount could increase average order value while making the purchase more convenient for the customer.

VI. MACHINE LEARNING MODELLING

RQ7: Can machine learning models predict sales revenue accurately?

A. Choosing Features

Before training the model, I have decided carefully about which features to include and which features to exclude. The selected feature include: `Price Each` and `Quantity Ordered`: had to be excluded. Since Sales is literally calculated as:

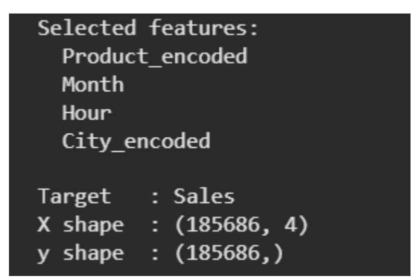
$$\text{Sales} = \text{Price Each} \times \text{Quantity Ordered} \tag{1}$$

including them would mean the model was just learning to multiply rather than discovering anything meaningful. This kind of data leakage produces artificially perfect results that would not hold up in a real prediction scenario.

The features I selected for building machine learning model were:

- `Product_encoded`: since product identity strongly determines price range
- `Month`: to capture seasonal purchasing patterns
- `Hour`: to capture time-of-day effects
- `City_encoded`: to capture any geographic differences in purchasing

Fig. 11 shows the selected features, target variable, and final dataset shape confirmed after encoding.



```
Selected features:
  Product_encoded
  Month
  Hour
  City_encoded

Target   : Sales
X shape  : (185686, 4)
y shape  : (185686,)
```

Fig. 11. Selected features, target variable, and dataset shape after encoding

`Product` and `City` were converted from text to integers using Label Encoding since machine learning models require numerical input.

B. Splitting the Data

I used a 70/30 train-test split with random_state = 42, giving 129,980 training rows and 55,706 test rows. The fixed random state means anyone running the notebook will get the same split.

C. Models Trained

I trained two models to compare:

Random Forest Regressor [4] builds many decision trees and averages their predictions. It handles non-linear patterns well and tends to be robust in practice. Given that sales in this dataset vary dramatically by product, I expected it to perform well.

Linear Regression was included as a baseline. It assumes a simple straight-line relationship between features and the target, which I suspected would struggle with a dataset where a single feature (product type) drives almost all the variance.

D. Model Evaluation

Table V compares the performance of both models on the test set.

TABLE V
MODEL PERFORMANCE COMPARISON

Model	$\mathbf{R^2}$	**RMSE**
Random Forest	0.9978	15.38
Linear Regression	0.0174	328.14

Random Forest explained 99.78% of the variance in Sales with an average prediction error of just 15.38 per order. Linear Regression barely managed an R^2 of 0.017, with errors averaging over 328 per order. The gap between the two models is substantial and tells us something important, the relationship between these features and Sales is fundamentally non-linear, and a straight line simply cannot capture it.

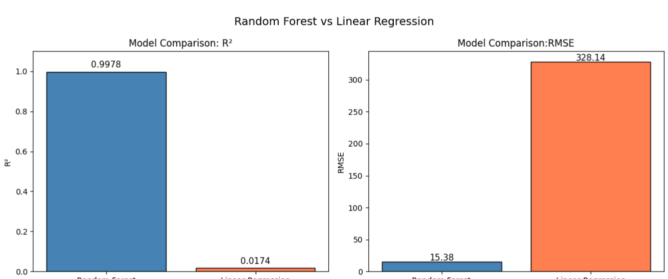


Fig. 12. R^2 and RMSE comparison between Random Forest and Linear Regression

Looking at the predicted vs true plot in Fig. 13, the Random Forest predictions track closely along the regression line across most of the price range. There are some deviations at the higher end, for example around the $1,700 MacBook Pro range, but overall the model is performing well.

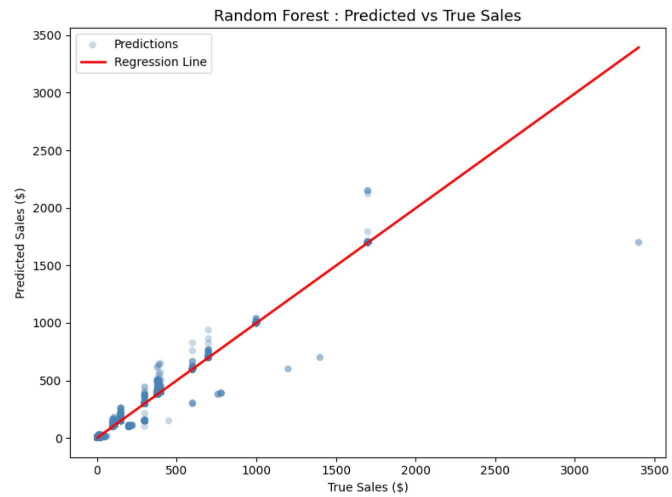


Fig. 13. Random Forest predicted vs true sales values on the test set

VII. FEATURE IMPORTANCE ANALYSIS

RQ8:Which features are most important in predicting sales?

Fig. 14 and Fig. 15 show the feature importance scores from the Random Forest model.

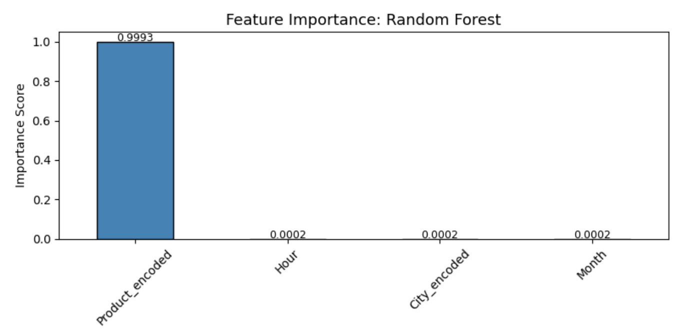


Fig. 14. Feature importance scores from the Random Forest model

```
Feature Importance Scores:
Feature                  Score    Percentage
--------------------------------------------
  Product_encoded       0.9993        99.9%
  Hour                  0.0002         0.0%
  City_encoded          0.0002         0.0%
  Month                 0.0002         0.0%
```

Fig. 15. Feature importance scores and percentages

The feature importance scores told a very clear story. Product identity accounted for 99.9% of the model's predictive power with an importance score of 0.9993, while Hour, City and Month each contributed just 0.02% , not much.

When stepping back, this result is not surprising. This is a fixed-price catalogue, every product has one price that never changes across orders. As soon as the model identifies the product, it already knows roughly what the sales value will be. The time of day, the city, and the month of the order add very little on top of that.

It is worth noting that Hour, City and Month were reasonable features to include, the EDA showed clear peak shopping hours and revenue differences across cities. However, once the model knew which product was ordered, none of that additional context made any meaningful difference.

This finding ties together everything else in the report. The Pearson correlation showed Price Each and Sales at r = 1.00. The Chi-Square test confirmed a strong association between Product and Sales (Chi-Square = 370,114, p = 0.000). The feature importance scores confirm it conclusively, product identity is the single dominant driver of sales revenue in this dataset.

VIII. Discussion

A. Summary of Key Findings

Working through this sales dataset gave me a much clearer picture of how sales patterns play out in electronic retail. A few insights stood out particularly:

December was clearly the strongest month, and the drop-off into January was sharp. This seasonal pattern is predictable but important, it means retailers need to plan their inventory and marketing well ahead of Q4.

San Francisco's position at the top of the city rankings was interesting. It is not the most populous city in the dataset, but its customers appear to spend more per order, likely because of a higher proportion of premium product purchases.

The co-purchase patterns between phones and cables were the clearest signal in the market basket analysis. A lift of 2.18 is a meaningful association, bundling these products or recommending cables at checkout for phone purchases seems like an obvious win.

The machine learning results were dominated by product identity, which reflects the nature of this particular dataset, fixed prices per product make prediction straightforward once you know what was ordered.

B. Recommendations

Based on the analysis, I would suggest:

- Introduce phone and cable bundles to increase average order value, the association rules make a strong case for this
- Ramp up promotional activity in November and early December to capture the holiday shopping surge
- Consider targeted marketing during the peak order hours identified in RQ4
- Focus higher inventory levels on San Francisco and Los Angeles given their revenue contribution

IX. Conclusion

This project walked through a full data science workflow on a real retail dataset, from cleaning and exploration through to machine learning and actionable insights. The analysis answered all eight research questions and uncovered patterns that have clear practical implications for how an electronic retailer could optimize its operations.

The Random Forest model performed exceptionally well (R^2 = 0.9978), though it is worth acknowledging that much of this performance comes from the fixed-price structure of the catalogue rather than complex learned patterns. In a real-world setting with more variable pricing or promotional discounts, the problem would be harder.

A. Limitations

- The dataset covers only year 2019, a single year of data will not be sufficient to represent longer-term trends or account for how shopping behaviour may have changed since then.
- Only nine US cities are included, which means the findings reflect a specific slice of the market and may not apply to other regions or countries.
- No customer demographic data was available. Knowing more about who is buying, age, income, or household type, would have allowed for deeper segmentation and more targeted insights.
- The Random Forest model's high R^2 of 0.9978 is largely a result of fixed product pricing rather than the model learning complex patterns. In a dataset where prices vary dynamically, performance would likely be considerably lower.

B. Future Work

If I were to extend this project:

- Bringing in multiple years of data to accurately predict the sales revenue at longer-term trends
- Adding customer demographic data to enable segmentation
- Experimenting time-series forecasting models to predict future monthly revenue
- Exploring clustering to group customers by purchasing behaviours

References

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